

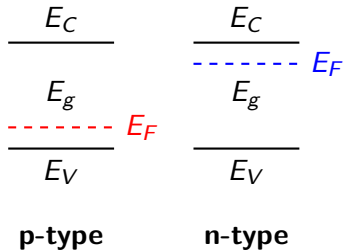
Lecture 18: final review

Prof. Daniel Schwalbe-Koda

MAT SCI 120 — Winter 2026

P-N junction: bringing two doped semiconductors together

- ▶ **p-type:** holes are majority carriers, Fermi level near valence band
- ▶ **n-type:** electrons are majority carriers, Fermi level near conduction band
- ▶ Contact creates interface where carriers must redistribute
- ▶ **Analogy with thermodynamics: Fermi level is a chemical potential!**
- ▶ **Analogy with kinetics: Fermi level is a concentration of electrons!**



Fermi level is a concentration of electrons: Fick's laws drive diffusion

- ▶ Carrier concentration set by Fermi level:

$$n = N_C \exp\left(\frac{E_F - E_C}{k_B T}\right)$$

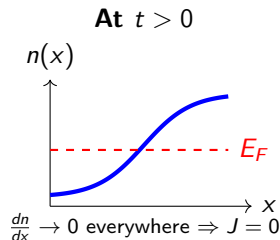
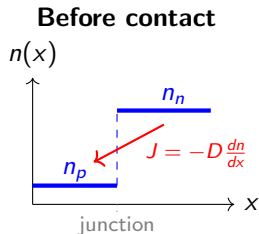
- ▶ Spatial variation in $E_F \Rightarrow \nabla n \neq 0$
- ▶ **Fick's 1st law** (flux $\propto$ gradient):

$$J = -D \frac{dn}{dx}$$

- ▶ **Fick's 2nd law** (evolution):

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$$

- ▶ Equilibrium: $\frac{dn}{dx} = 0 \Leftrightarrow E_F = \text{const}$



Poisson equation in depletion region

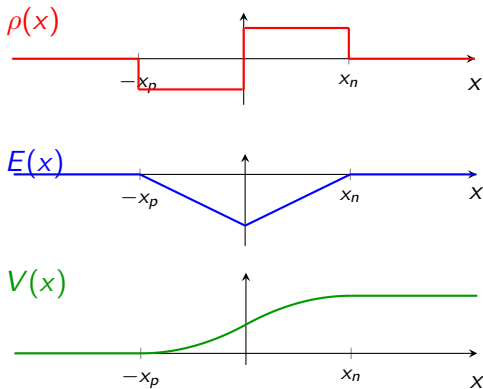
- Charge density in depletion approximation:

$$\rho(x) = \begin{cases} -eN_A & -x_p < x < 0 \\ +eN_D & 0 < x < x_n \end{cases}$$

- Poisson equation:

$$\frac{d^2 V}{dx^2} = -\frac{\rho}{\epsilon}$$

- Integrate once $\Rightarrow E(x)$ linear
- Integrate twice $\Rightarrow V(x)$ quadratic
- Boundary conditions: $E = 0$ and V continuous at $\pm x_{p,n}$



Built-in voltage from thermodynamic argument

Combine the Boltzmann relation with the concentration ratio:

$$\frac{n_n}{n_p} = \frac{N_A N_D}{n_i^2} = \exp\left(\frac{eV_{bi}}{k_B T}\right)$$

Solve for V_{bi} :

$$V_{bi} = \frac{k_B T}{e} \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

This thermodynamic expression must equal the electrostatic result:

$$V_{bi} = \frac{e}{2\epsilon} (N_A x_p^2 + N_D x_n^2)$$

Equating the two gives a relation between the depletion widths x_p , x_n and the doping concentrations N_A , N_D .

Shockley diode equation: derivation

Four-step derivation:

1. **Equilibrium:** Boltzmann $\Rightarrow$

$$I_{\text{diff}}^0 \propto e^{-eV_{bi}/k_B T} = I_{\text{drift}} \equiv I_0$$

2. **Under bias** U_1 : barrier $\rightarrow e(V_{bi} - U_1)$

$$I_{\text{diff}} = I_0 \exp\left(\frac{eU_1}{k_B T}\right)$$

3. **Drift is bias-independent:** minority carriers swept by field $\Rightarrow I_{\text{drift}} \approx I_0$

4. **Net current:**

Shockley diode equation

$$I = I_0 \left[\exp\left(\frac{eU_1}{k_B T}\right) - 1 \right]$$

$$I_0 \propto n_i^2 \propto \exp\left(-\frac{E_g}{k_B T}\right)$$

Thermal voltage

At 300 K: $k_B T/e \approx 26 \text{ mV}$

Si turn-on $\approx 0.6\text{--}0.7 \text{ V}$

Shockley diode equation: limiting regimes

Three regimes:

A — Forward bias ($eU_1 \gg k_B T$):

$$I \approx I_0 e^{eU_1/k_B T}$$

Exponential turn-on; current rises steeply.

B — Reverse bias ($eU_1 \ll -k_B T$):

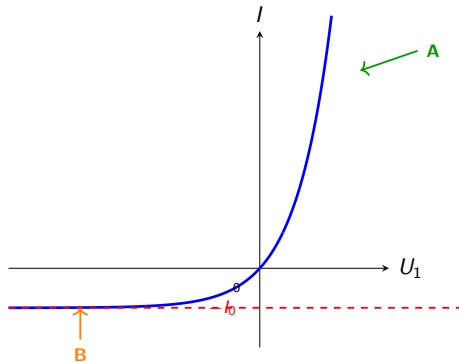
$$I \approx -I_0$$

Saturates at small reverse current.

C — Temperature dependence:

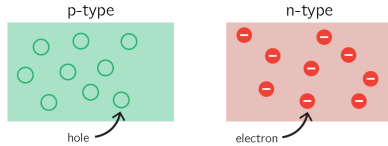
$$I_0 \propto e^{-E_g/k_B T}$$

Strong T -dependence through n_i^2 .

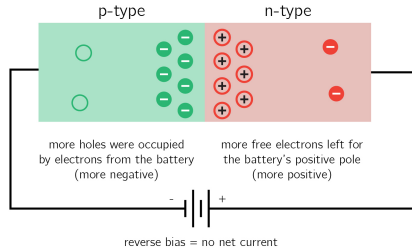


Summary of P-N junctions

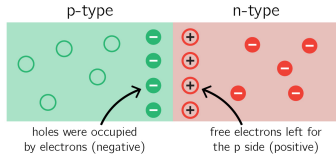
prior to contact



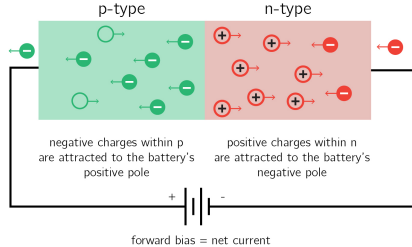
reverse bias



after contact



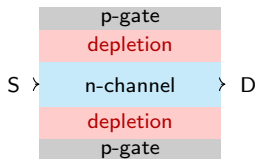
forward bias



J-FET vs. MOSFET: operating principles and pinch-off conditions

J-FET (depletion mode)

- ▶ Gate reverse-biases p-n junction $\Rightarrow$ widens depletion region $\Rightarrow$ **narrows existing** n-channel
- ▶ Channel exists at $V_{GS} = 0$; device is *normally on*
- ▶ Pinch-off: depletion closes channel completely
- ▶ $I_{DS} \rightarrow 0$ when $|V_{GS}| \geq |V_P|$ (pinch-off voltage)

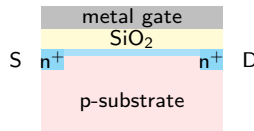


MOSFET (enhancement mode)

- ▶ Gate field across SiO_2 attracts minority carriers $\Rightarrow$ **creates inversion layer** (channel did not exist before)
- ▶ Channel appears only when $V_{GS} > V_T$; device is *normally off*
- ▶ Pinch-off at drain end when:

$$V_{DS,\text{sat}} = V_{GS} - V_T$$

- ▶ Beyond pinch-off: I_{DS} saturates



Photon absorption thresholds and Beer–Lambert attenuation

Absorption threshold by material class:

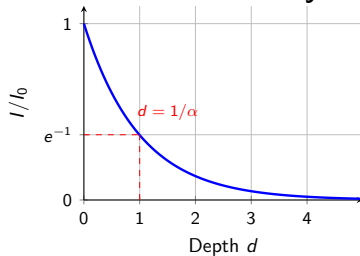
- ▶ **Metals:** continuous empty states above $E_F \Rightarrow$ absorb any $h\nu$; no threshold $\Rightarrow$ opaque & reflective
- ▶ **Semiconductors:** threshold at $h\nu = E_g$; transparent for $h\nu < E_g$
- ▶ **Insulators:** $E_g > 2.5 \text{ eV} \Rightarrow$ absorption edge in UV $\Rightarrow$ transparent in visible
- ▶ **Defects** create sub-gap states $\Rightarrow$ selective absorption (e.g., Cr^{3+} in $\text{Al}_2\text{O}_3 \Rightarrow$ ruby red)

Beer–Lambert law:

$$I(d) = I_0 e^{-\alpha d}$$

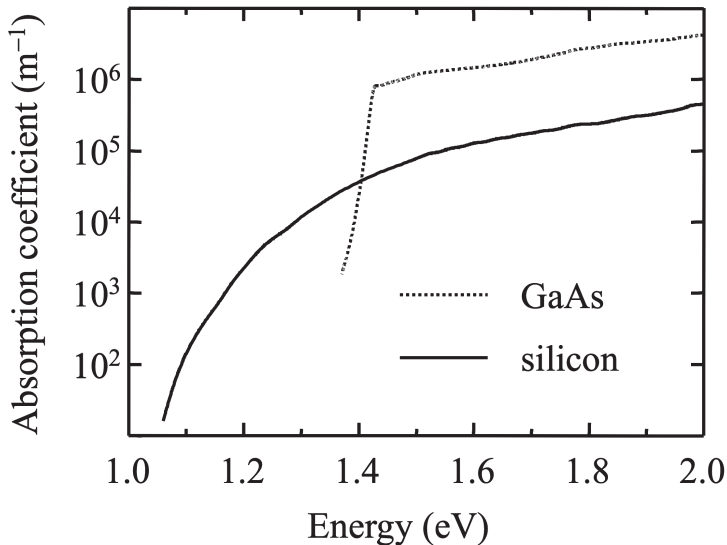
- ▶ α : absorption coefficient [cm^{-1}], material- and λ -dependent
- ▶ Penetration depth $\delta = 1/\alpha$

Beer–Lambert decay



Dependence of absorbance with band gap type (direct vs. indirect)

- ▶ GaAs: direct band gap
- ▶ Si: indirect band gap
- ▶ absorption of GaAs is higher and grows sharply



Band gap $\rightarrow$ color $\rightarrow$ alloying $\rightarrow$ LED design: the optoelectronics synthesis chain

The synthesis chain:

1. **Band gap $\rightarrow$ color:** photons with $h\nu \geq E_g$ absorbed; emitted photon has $h\nu \approx E_g$
2. **LED:** forward-biased junction $\Rightarrow$ carrier injection $\Rightarrow$ radiative recombination; *direct gap required* for efficiency

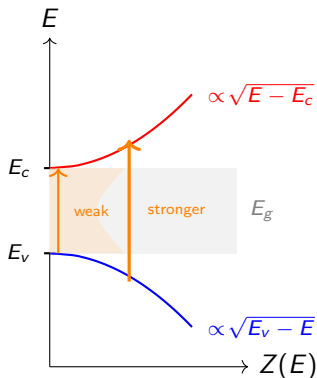
Example: $\text{In}_x\text{Ga}_{1-x}\text{N}$ system

- ▶ GaN: $E_g \approx 3.4 \text{ eV}$ (UV)
- ▶ InN: $E_g \approx 0.7 \text{ eV}$ (IR)
- ▶ Tunable across **entire visible spectrum**

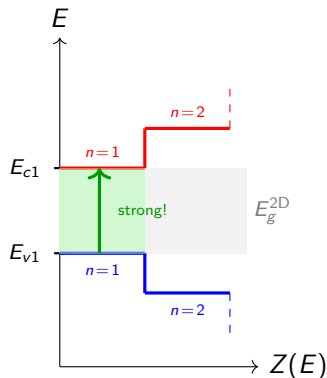
Color	E_g (eV)	λ (nm)
Red	1.8	690
Green	2.3	540
Blue	2.7	460
UV	3.4	365

DOS and photon absorption

3D (bulk)



2D (quantum well)

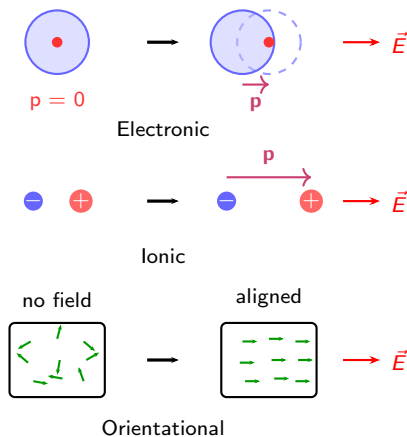


DOS in 2D materials is a sharp step → **higher probability of transitions near for photons near E_g** → emission is more well-defined!

Quantum-well LEDs and laser diodes exploit the 2D DOS to concentrate emission at a single well-defined energy, improving efficiency and spectral purity.

Three polarization mechanisms in dielectrics

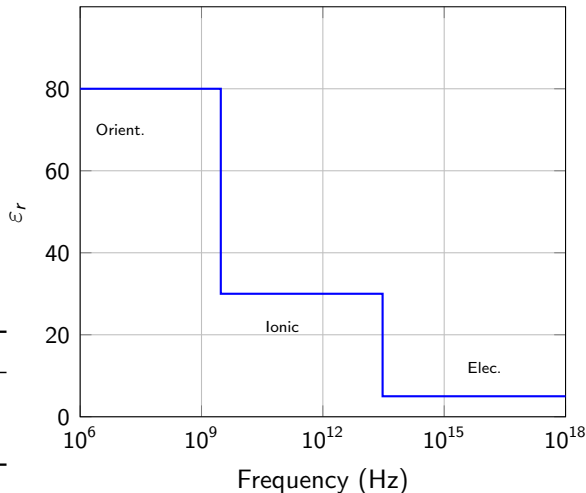
- ▶ **Electronic:** electron cloud displacement (all materials)
- ▶ **Molecular (ionic):** ion displacement in ionic crystals
- ▶ **Orientalional:** permanent dipole alignment (polar molecules)
- ▶ Each has characteristic frequency cutoff



Frequency dependence of dielectric constant

- ▶ ϵ_r decreases stepwise as mechanisms freeze out
- ▶ Orientational lost above $\sim 10^9$ Hz (microwave)
- ▶ Ionic lost above $\sim 10^{13}$ Hz (infrared)
- ▶ Only electronic remains at optical frequencies

Mechanism	Timescale	Freeze-out	Example
Orientational	ps–ns	$\sim 10^9$ Hz	H ₂ O
Ionic	~ 0.1 ps	$\sim 10^{13}$ Hz	NaCl
Electronic	$\sim$ fs	$\sim 10^{15}$ Hz	all



Diamagnetism vs paramagnetism

- ▶ **Why is diamagnetism universal?**

All electrons have orbital motion

- ▶ **Why is diamagnetism weak?**

Induced moments $\ll$ permanent moments

- ▶ Key concept: induced (dia) vs pre-existing aligned (para)

	Dia	Para
χ_m sign	< 0	> 0
$ \chi_m $	$\sim 10^{-5}$	$\sim 10^{-3}$
Origin	induced	pre-existing
Electrons	paired	unpaired

Worked examples:

- ▶ Fe^{3+} ($3d^5$, 5 unpaired e^-) $\Rightarrow$ paramagnetic (or ferromagnetic in bulk Fe)
- ▶ Zn^{2+} ($3d^{10}$, 0 unpaired e^-) $\Rightarrow$ diamagnetic

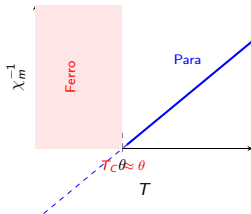
$$B = \mu_0(1 + \chi_m)H$$

Curie temperature, domain formation, and the origin of hysteresis

Curie–Weiss law ($T > \theta$):

$$\chi_m = \frac{C}{T - \theta}$$

- ▶ $\chi_m \rightarrow \infty$ as $T \rightarrow \theta^+$: zero field can magnetize the material
- ▶ Below T_C : spontaneous magnetization
- ▶ Above T_C : material is paramagnetic (needs H to magnetize)



Domain formation as energy minimization:

- ▶ Single-domain: high magnetostatic (stray-field) energy, low entropy
- ▶ Multi-domain: reduces stray field; closure domains eliminate it entirely
- ▶ Domain walls cost exchange + anisotropy energy $\Rightarrow$ equilibrium domain size

Origin of hysteresis — domain-wall pinning:

- ▶ Defects (dislocations, grain boundaries) pin domain walls